Adapted Weights, Inherited Tokenizer: Measuring the Token Cost of Hausa, Igbo and Yoruba in N-ATLaS

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Abstract

A model can be adapted for a language without changing how that language is tokenized. In this study we examine N-ATLaS, a Llama-based model fine-tuned for English, Hausa, Igbo and Yoruba, and ask whether its adaptation changed how those languages are tokenized. We find no change to the vocabulary or to how ordinary text is split. At pinned revisions, N-ATLaS has the same 128,256-entry token-to-ID mapping and the same loaded tokenizer as Llama 3.1. It shares the chat template of Llama 3.1 Instruct, and the one difference we found among the inspected special-token roles is an assigned padding token. On 2,009 aligned FLORES+ sentences in NFC (Unicode Normalization Form C), the same content takes 2.02 times the English tokens in Hausa, 2.37 in Igbo and 2.69 in Yoruba. Controlled spelling changes show where that excess sits. Removing combining marks eliminates 58.7% of the measured Yoruba excess and 25.1% of the Igbo excess, and replacing Hausa's hooked letters eliminates 10.5%. These are effects of lossy perturbations, not causal shares. Unicode form matters as well. Canonically equivalent Yoruba text in NFD (Unicode Normalization Form D) needs 18.7% more tokens than NFC under N-ATLaS and 26.5% more under Gemma 4, which roughly halves Gemma 4's advantage. Gemma 4 uses 9.2% to 17.0% fewer tokens than N-ATLaS across translated text, news and tweets, with vocabulary size and tokenizer design confounded. We measure token counts only, not model quality, latency or price. The results support auditing tokenizer inheritance and input normalization whenever a model is adapted for Nigerian languages.

1 Introduction

N-ATLaS was released in September 2025 as a language model for English, Hausa, Igbo and Yoruba. Its model card describes a Llama-based model with eight billion parameters, fine-tuned on about 392 million tokens of instruction data across the four languages, and developed by Awarri Technologies in partnership with the Federal Government of Nigeria (Awarri Technologies, 2025). A model built for these three Nigerian languages raises a simple question. Did the way the model splits them into tokens change?

The question matters because a language model never sees text directly. A tokenizer first splits the text into units from a fixed vocabulary, and the number of units decides how much of a passage fits in a context window. Tokenizers do not treat languages equally. The same content can take several times more tokens in one language than in another (Petrov et al., 2023), and speakers of many languages pay more for commercial models while getting worse results (Ahia et al., 2023). For African languages the gap is now well documented (Lundin et al., 2026; Somide, 2026).

Fine-tuning changes a model's weights. It does not have to change the tokenizer, and when the tokenizer stays the same, every input is split exactly as it was before. So a model can get better at a language and still represent that language at the old cost. One community evaluation reports that N-ATLaS does

outperform Llama-3-8B-Instruct on Hausa, Igbo and Yoruba tasks (Ajayi and Oladipo, 2025). Our study neither tests this nor depends on it, but the released files do let us test the other half of the story: whether the tokenizer moved.

We study the tokenizer distributed with N-ATLaS at a fixed repository revision, using tokenizer files only. We first audit what it shares with the Llama releases, separating four things that are easy to blur: byte-level file identity, vocabulary identity, identity of the loaded tokenizer, and identical outputs on real text. We then measure how many tokens Hausa, Igbo and Yoruba need relative to English on aligned translations, and test how sensitive those counts are to tone marks, underdotted letters, Hausa's hooked letters and Unicode normalization form. Finally, we compare N-ATLaS with Gemma 4 on news and tweets to see whether the difference between the two tokenizers holds outside translated text.

We find no change to the vocabulary or to how ordinary text is split. N-ATLaS has the complete vocabulary of Llama 3.1, and the same content costs 2.02 times the English tokens in Hausa, 2.37 in Igbo and 2.69 in Yoruba. Much of the Yoruba excess is tied to how the language is correctly written, and the Unicode form of the text alone changes Yoruba token counts by almost a fifth.

The paper makes four contributions.

1. **A tokenizer audit of a Nigerian language model.** We show that N-ATLaS shares its full token-to-ID mapping and loaded tokenizer with Llama 3.1, and its chat template with Llama 3.1 Instruct. The audit also corrects our own first reading, which compared against Llama 3 and mistook Llama 3.1's control tokens for changes made to N-ATLaS.
2. **Paired measurements of the token premium.** We report token-total and fertility premiums over English with confidence intervals, for N-ATLaS and for Gemma 4.
3. **Controlled orthographic and normalization experiments.** We separate tone marks, underdotted letters and hooked letters, and we isolate the effect of NFC against NFD.
4. **Replication across three kinds of text.** The difference between the two tokenizers holds in translated benchmark text, news and tweets.

We do not evaluate what N-ATLaS can do, and we do not identify the checkpoint its weights were trained from. Our claims are about representation and token counts.

2 Related work

Tokenization inequality across languages. That tokenizers serve languages unequally is well established. Rust et al. (2021) compared multilingual and monolingual models across nine languages and found that a dedicated monolingual tokenizer matters as much as pretraining data size for downstream performance. Petrov et al. (2023) used translations of the same text to show that tokenized length can differ by up to 15 times across languages, with consequences for cost, latency and context. Ahia et al. (2023) studied commercial APIs across 22 typologically diverse languages and found that speakers of many languages are charged more while receiving worse results. These studies are the reason we measure representation length separately from model performance.

Later work asks how far tokenizer measures explain model quality. Tsvetkov and Kipnis (2024) propose Information Parity and report that it correlates with benchmark scores better than tokenizer parity or fertility. Chiu (2026) evaluates 24 tokenizers across 15 languages on six intrinsic metrics and finds that they predict per-language benchmark performance, while noting that the evidence is correlational. Ovcharov (2026) measures ten models across 25 European languages and reports that fertility rankings hold across three text registers.

The tokenizer tax on African languages. Recent work calls the extra tokens a language needs over English a tax (Lundin et al., 2026; Ovcharov, 2026; Somide, 2026). We use premium for the ratio we

measure, and both words refer to the same extra cost. Two recent studies give the closest context for ours. Lundin et al. (2026) evaluate ten models on AfriMMLU across 16 African languages and find that higher token fertility predicts lower accuracy across models and subjects. Somide (2026) measures 20 African languages, including Hausa, Igbo and Yoruba, on current tokenizers that include Llama 3.1 and Gemma 4. It finds a premium over English for every language and reports Gemma 4 as the most efficient tokenizer on average, with no tokenizer removing the penalty. That study defines premium as fertility relative to English and normalizes text to NFC before counting.

Our fertility measurements agree closely with Somide's (Section 4.2), and we do not claim that the disparity itself is a new finding. Our primary metric, the token-total premium, is a different quantity from a fertility premium. Our focus is also different. We ask what a model adapted for these languages inherited, and we add controlled spelling experiments and an NFC and NFD comparison for these three languages. Somide (2026) describes an NFC and NFD control for Ethiopic and Arabic-script text, not for Latin-script text.

Adapting models to African languages. Recent African-language models are built by adapting existing ones. Buzaaba et al. (2025) adapt large language models to African languages and find that mixing curated African-language data with English educational text improves performance. Yu et al. (2026) apply continued pretraining on 26 billion tokens to five base models for 20 African languages, including Hausa, Igbo and Yoruba, and identify data composition as the main driver of gains. Both evaluate task performance. To our knowledge, neither reports what the adapted model's tokenizer costs the target languages.

Methods for changing the tokenizer exist. Yamaguchi et al. (2024) compare five cross-lingual vocabulary adaptation methods on four generative models and report inference speedups of up to 271.5%. MAGNET (Ahia et al., 2024) and MYTE (Limisiewicz et al., 2024) propose tokenization schemes designed to reduce segmentation disparities across languages and scripts.

Closest to the adaptation setting we study, Purason et al. (2026) propose continued BPE training, which extends a pretrained tokenizer by continuing merge learning on new data. They report better tokenization efficiency across several languages and model families.

Orthography and diacritics. Marked orthography is a known source of fragmentation. Inoue et al. (2026) show that full Arabic diacritization substantially increases token fragmentation and degrades benchmark performance, while models are robust to the limited diacritics found in natural text. For Yoruba, Adelani et al. (2021) built a corpus with curated orthography and examined how diacritics affect Yoruba and English machine translation. Nigerian Latin-script orthographies add two things to this picture. Underdotted characters are letters of the alphabet, not optional marks, and Hausa's hooked letters cannot be reached by removing decomposable marks at all.

The gap. Prior work establishes that African languages pay a tokenizer premium. It does not ask what happens to that premium when a model is adapted specifically for the languages that pay it. We take N-ATLaS as a case, audit what its adaptation left unchanged in the tokenizer, measure the premium that remains, and test how much of it moves with each language's orthography and with Unicode normalization.

3 Methods

3.1 Questions and scope

We ask four questions.

1. Which tokenizer components does N-ATLaS share with the Llama releases?
2. How many tokens do Hausa, Igbo and Yoruba need relative to aligned English text?

3. How sensitive are those counts to specific orthographic changes and to Unicode normalization form?
4. Does Gemma 4's advantage over N-ATLaS hold across translated text, news and tweets?

Every comparison uses pinned tokenizer revisions and `add_special_tokens=False`, so no tokenizer adds control tokens that the others lack. We load no model weights and apply no chat template when counting corpus text. Full revisions and file hashes are in Appendix A.

Table 1. Tokenizer releases and their role in the study.

Tokenizer	Repository	Vocabulary entries	Role
N-ATLaS	NCAIR1/N-ATLaS	128,256	Subject
Llama 3 8B	meta-llama/ Meta-Llama-3-8B	128,256	Base named on the N-ATLaS model card
Llama 3.1 8B	meta-llama/Llama-3.1-8B	128,256	Full mapping audit
Llama 3.1 8B Instruct	meta-llama/ Llama-3.1-8B-Instruct	128,256	Mapping and chat template audit
Gemma 4 12B	google/gemma-4-12B	262,144	Alternative tokenizer

We include Gemma 4 because we wanted a second, separately trained tokenizer to compare against, and Somide (2026) reports it as the most efficient one on average across the African languages that study measures. It is not the most efficient for our three languages in that study, where o200k_base has a lower premium for all three and BLOOM for Igbo and Yoruba. We did not measure those tokenizers. Its vocabulary is about twice the size of N-ATLaS's. The comparison therefore does not hold vocabulary capacity or training data constant, and we return to this in Section 6.

3.2 Identity audit

"Same tokenizer" can mean several things, so we check each one separately. At the file level, we hash `tokenizer.json`, `tokenizer_config.json` and `special_tokens_map.json` with SHA-256. At the component level, we compare every token string and its ID, including the control-token range, along with the BPE merges, normalizer, pre-tokenizer, decoder and post-processor. A canonical serialization of the loaded tokenizer backend gives a further check on the components together. We also inspect the BOS, EOS and PAD assignments and compare the chat template strings exactly.

At the behavioural level, we compare complete token-ID sequences on unchanged corpus text. Equal token counts are not enough; the sequences of IDs must match. N-ATLaS and Llama 3 are compared on all three corpora, 88,467 texts in total. The Llama 3.1 base and Instruct comparisons cover the 8,036 FLORES+ texts.

3.3 Corpora and preparation

Table 2. Prepared corpus sizes (sentences or tweets per language).

Corpus	Text type	English	Hausa	Igbo	Yoruba
FLORES+ (NLLB Team, 2024)	Aligned translations	2,009	2,009	2,009	2,009
MasakhaNER 2.0 (Adelani et al., 2022)	News	none	7,913	10,381	9,633
NaijaSenti (Muhammad et al., 2022)	Tweets	none	21,937	15,677	14,890

FLORES+ release 4.6 is the primary corpus: 997 dev and 1,012 devtest sentences per language, in Latin script. It is maintained by the Open Language Data Initiative (Open Language Data Initiative, n.d.; Burchell et al., 2024; Dale et al., 2025) and builds on the earlier FLORES benchmarks (Guzmán et al., 2019; Goyal et al., 2022; NLLB Team, 2024). The dataset credits Abdulmumin et al. (2024) for improvements to its Hausa data. The same 2,009 sentence IDs exist in English, Hausa, Igbo and Yoruba. We need this alignment because an English-relative premium only means something when the content is held fixed.

The other two corpora test whether the comparison between tokenizers survives outside translated text. For MasakhaNER 2.0, we rebuild each sentence by joining the published CoNLL tokens with a single ASCII space. Punctuation stays separately spaced, so these strings do not reproduce the original article typography. We remove exact duplicates within each language across splits: 252 Hausa, 524 Igbo and 190 Yoruba sentences. For NaijaSenti, we keep the released tweet strings and remove exact duplicates in the same way: 215 Hausa, 38 Igbo and 237 Yoruba tweets. We also evaluate the tweets after removing HTTP(S) URLs and user mentions. Task labels are not used, and all available splits contribute.

Neither natural corpus has an aligned English version, so their results compare two tokenizers on the same text and do not estimate an English-relative premium. Released tweets were pre-processed before release and may contain code-mixing. We do not treat a language label as proof that a text is monolingual.

The FLORES+ metadata records a CC BY-SA 4.0 licence. For MasakhaNER we follow the original repository's CC BY-NC 4.0 statement; its Hugging Face header says AFL-3.0, which conflicts with that statement. NaijaSenti records CC BY-NC-SA 4.0. We keep these source statements and do not publish corpus text.

3.4 Normalization and metrics

Unicode can store the same text in more than one way. In Normalization Form C (NFC), a letter and its marks are combined into a single code point wherever one exists. In Normalization Form D (NFD), they are stored separately, as a plain letter followed by its combining marks. The Yoruba word ẹ̀kọ́ is five code points in NFC and seven in NFD (Figure 1). The two forms look identical on screen and count as the same text under Unicode's rules (Unicode Consortium, 2026). A tokenizer, however, receives a different sequence of code points for each, and can split them differently.



Figure 1. The same Yoruba word in NFC and in NFD. Each box is one code point. Both forms display identically, but they give the tokenizer different input.

Published FLORES+ text is not uniformly in NFC. As published, 576 Yoruba and 69 Igbo sentences differ

from their NFC forms, while all Hausa and all English sentences are already NFC. We use NFC input for the English-relative premiums, the orthographic experiments, characters per token and FLORES+ fragmentation. Without a common reference, a change in normalization would be counted as an effect of removing marks. The cross-corpus comparison in Section 4.5 uses released or reconstructed text with no extra normalization, and we label that difference where it applies.

Let T be the total tokens for a language over the aligned sentences and W its total whitespace-delimited units. Our primary measure is the **token-total premium**:

$$P_L^{\text{tok}} = \frac{T_L}{T_{\text{English}}} \tag{1}$$

We make it primary because it answers the question a user would ask, which is what the same content costs, and because it needs no definition of a word. **Fertility** is tokens per whitespace unit, and the secondary **fertility premium** divides a language's fertility by English fertility:

$$F_L = \frac{T_L}{W_L}, \qquad P_L^{\text{fert}} = \frac{F_L}{F_{\text{English}}} = P_L^{\text{tok}} \cdot \frac{W_{\text{English}}}{W_L} \tag{2}$$

The two premiums answer different questions and are not interchangeable. We report the fertility premium because it is the definition used by Somide (2026).

For a corpus with token-total premium P, the reciprocal 1/P describes how much content fits in a fixed token budget relative to English, ignoring formatting overhead. This is a corpus-level approximation and not a prediction for every prompt. We report no price, latency or energy multiplier.

Characters per token counts Unicode code points, including whitespace. Fragmentation is the share of whitespace units overlapped by one, two, three, or four or more tokenizer offset spans, with punctuation attached to its unit. As a sensitivity check on the unit itself, we repeat the fertility premium using `regex` word boundaries and keep only segments that contain letters or digits. This is a boundary diagnostic and not a validated linguistic segmentation.

For two tokenizers on identical text, Gemma 4's relative reduction is:

$$R = 1 - \frac{T_{\text{Gemma}}}{T_{\text{N-ATLaS}}} \tag{3}$$

For a perturbation that moves a premium from P before to P after, the fraction of measured excess removed is:

$$S = \frac{P_{\text{before}} - P_{\text{after}}}{P_{\text{before}} - 1} \tag{4}$$

S summarizes how a metric changes when the text is altered. It is not a causal decomposition of the language or of the tokenizer's design.

3.5 Orthographic experiments

Every altered sentence is compared with its own NFC reference.

- **Tone marks removed (Igbo, Yoruba).** Decompose to NFD, delete U+0300, U+0301 and U+0304, and recompose to NFC.
- **Underdots removed (Igbo, Yoruba).** Delete U+0323. U+0329 is inventoried and tested in an auxiliary condition.

- **Tone marks and underdots removed together.** Tested separately from the next condition, because the corpus contains a few other marks.
- **All marks removed.** Delete every nonspacing mark, Unicode category Mn.
- **Hooked letters replaced (Hausa).** Map ƙ, ɗ, ɓ and ƴ, with their uppercase forms, to k, d, b and y.
- **NFD encoding (Hausa, Igbo, Yoruba).** Convert the NFC text to NFD.

Figure 2 shows the letters involved in each language and what each alteration does to a real word.



Figure 2. The marked letters studied in each language, example words from FLORES+ with those letters highlighted, and the effect of each alteration on one word.

NFC and NFD carry the same canonical text and differ only in the code-point sequence the tokenizer receives (Unicode Consortium, 2026). The other alterations discard real distinctions. Removing underdots changes ṣ as well as vowels, and replacing hooked letters merges distinct Hausa consonants. We use them to see how the tokenizers respond, and they are not recommendations for how anyone should spell.

Two further diagnostics support the experiments. We analyse FLORES+ word types, defined as runs of Unicode letters and marks with internal apostrophes or hyphens, normalized to NFC and casefolded; types that occur at least five times are ranked by tokens per occurrence and by total excess tokens. We also decode every vocabulary entry on its own and count the entries that contain a complete target lowercase letter. Entries holding only part of a multi-byte character decode to a replacement character. This audit describes how the vocabulary is allocated. It does not measure whether a letter can be represented.

3.6 Uncertainty and significance

All confidence intervals are 95% percentile intervals from 10,000 bootstrap resamples. Resampling keeps the pairing each comparison needs: aligned English and target sentences for premiums, unchanged and altered versions for perturbations, and the same texts under both tokenizers for relative reductions. The implementations do not share one global set of draws across every language and tokenizer. Seed 20260930 is used for the premium, refinement and cross-corpus analyses, and 20261001 for the focused orthography run.

We also run two-sided Wilcoxon signed-rank tests on paired token counts, using the approximate method with `zero_method='zsplit'`. Holm correction covers four languages for the FLORES+ tokenizer comparison, three languages for each natural corpus, and all 34 rows of the focused orthography summary. With about 2,000 pairs per comparison, small differences reach significance, so effect sizes and intervals carry the interpretation and the tests are reported in Appendix C.

The news and tweet releases carry no article, author or thread identifiers. Their bootstraps therefore treat texts as independent, and dependence among sentences or tweets may make those intervals too narrow.

4 Results

4.1 N-ATLaS carries the complete Llama 3.1 vocabulary

We started with Llama 3, because that is the base the model card names. The lexical part of the comparison was clean. N-ATLaS and Llama 3 have the same token-to-ID mapping below ID 128,000 and the same merges and text-processing components, and they return identical token-ID sequences on all 8,036 FLORES+ texts, 27,927 news sentences and 52,504 tweets. The control-token range did not match. Three token strings were unique to each side, and 246 shared strings had different IDs. Read on its own, that looks like N-ATLaS changing how control tokens are handled.

It was not. The three strings found only in N-ATLaS are special tokens that Meta defines for Llama 3.1 (Meta, n.d.), so we repeated the audit against the Llama 3.1 base and Instruct releases. N-ATLaS shares the entire 128,256-entry token-to-ID mapping with both. There are no unique token strings and no mismatched IDs. The canonical loaded tokenizer definitions match as well, including merges and text-processing components, and all 8,036 FLORES+ texts return identical ID sequences in both comparisons. The control-token discrepancy with Llama 3 is simply the difference between Llama 3 and Llama 3.1. We found no evidence of any vocabulary change made for Hausa, Igbo or Yoruba.

Table 3. Identity findings relative to N-ATLaS.

Property	**Llama 3 8B**	**Llama 3.1 8B**	**Llama 3.1 8B Instruct**
Complete token-to-ID mapping	Differs in the control range	Identical	Identical
Lexical mapping below ID 128,000	Identical	Identical	Identical
Canonical loaded tokenizer	Differs in added tokens	Identical	Identical
Three source file hashes	All differ	All differ	All differ
BOS, EOS and PAD assignments	Differ	Differ	Only PAD differs
Chat template	Not audited	Absent	Identical
FLORES+ texts with matching IDs	8,036 of 8,036	8,036 of 8,036	8,036 of 8,036

The closest match is the Instruct release. N-ATLaS has the same BOS and EOS assignments and exactly the same chat template string. Among the BOS, EOS and PAD roles, the one difference is that N-ATLaS sets PAD to <|eot_id|>, where Instruct leaves it unset. The base release uses <|end_of_text|> as EOS, leaves PAD unset, and has no chat template at its pinned revision.

Two cautions apply. The files are not byte-identical: all three SHA-256 hashes differ in both comparisons, even though the files load to the same tokenizer. Our findings therefore establish that the audited encoding components are equivalent. They do not establish that padding is the only metadata difference of any kind.

The model configuration points the same way as the tokenizer. N-ATLaS declares `max_position_embeddings=131072` with `llama3` RoPE scaling, an original length of 8,192 and a factor of 8.0. Llama 3 declares 8,192 and no scaling. This is consistent with a Llama 3.1 style configuration. Neither that

resemblance nor the tokenizer equivalence tells us which checkpoint the N-ATLaS weights were trained from, or what its usable context length is in practice.

4.2 Aligned text needs two to 2.7 times the English tokens

On NFC FLORES+ text, N-ATLaS needs 2.021 times the English tokens for Hausa, 2.374 for Igbo and 2.689 for Yoruba (Table 4, Figure 3). Taking reciprocals, a fixed token budget holds about 49.5% as much Hausa content as English, 42.1% as much Igbo and 37.2% as much Yoruba, at corpus level.

Table 4. NFC token totals and token-total premiums, with 95% intervals.

Language	N-ATLaS tokens	N-ATLaS premium	Gemma 4 tokens	Gemma 4 premium
English	52,963	1.000	52,995	1.000
Hausa	107,034	2.021 [2.004, 2.038]	92,809	1.751 [1.737, 1.766]
Igbo	125,722	2.374 [2.351, 2.397]	113,399	2.140 [2.120, 2.160]
Yoruba	142,409	2.689 [2.654, 2.724]	125,611	2.370 [2.340, 2.400]

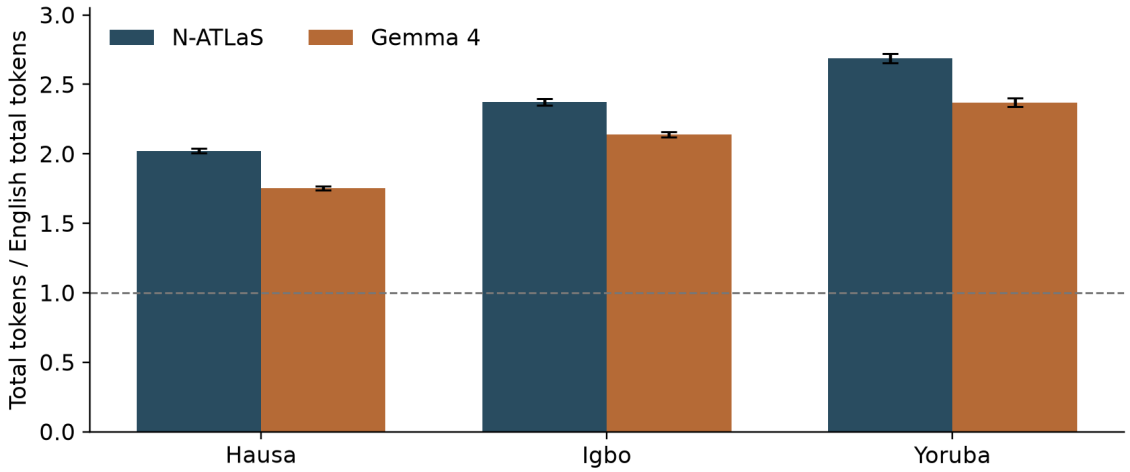


Figure 3. NFC token-total premiums on 2,009 aligned sentences per language. Error bars are paired-bootstrap 95% intervals. English equals 1 by definition. The figure shows token totals, not fertility ratios.

Gemma 4 uses 13.29% fewer Hausa tokens, 9.80% fewer Igbo tokens and 11.80% fewer Yoruba tokens on the same NFC text, and the two tokenizers are almost identical on English. Its premiums are lower but still large: 1.751, 2.140 and 2.370.

The secondary fertility premiums under N-ATLaS are 1.751 for Hausa, 2.103 for Igbo and 2.326 for Yoruba. They are smaller than the token-total premiums because the translations contain more whitespace units than the English source. Replacing whitespace units with the Unicode boundary measure gives 1.753, 2.045 and 2.334, which keeps the ordering of the languages. Appendix B reports both tokenizers. These values can be checked against Somide (2026), who reports fertility premiums on the devtest split of FLORES-200+ with Unicode word segmentation. For Llama 3.1, which tokenizes this text exactly as N-ATLaS does (Section 4.1), that study gives 1.718 for Hausa, 2.051 for Igbo and 2.351 for Yoruba, against our 1.753, 2.045 and 2.334 with the Unicode boundary measure. For Gemma 4 it gives 1.507, 1.851 and 2.073, against our 1.519, 1.844 and 2.057. The two studies used different code and different subsets of the corpus, and no pair of values differs by more than 0.04.

The same pattern shows at the level of individual words. Under N-ATLaS, 82.9% of English whitespace units map to a single token, against 34.0% for Hausa, 23.9% for Igbo and 26.9% for Yoruba. The share that needs four or more tokens rises from 1.1% in English to 10.7%, 19.6% and 27.8% (Appendix B, Table B3).

Figure 4 shows what this looks like in a single sentence. In English, each of the six words is one N-ATLaS token and the full stop is the seventh. The same sentence takes 14 tokens in Hausa, 16 in Igbo and 12 in Yoruba, because the words are cut into smaller pieces. One sentence cannot stand for the corpus, though. Here Yoruba costs less than Hausa, while across all 2,009 sentences it costs the most (Table 4).



Figure 4. One aligned FLORES+ sentence (dev, ID 212; CC BY-SA 4.0) in the four languages, as NFC text. Each box is one N-ATLaS token. The numbers on the right are the token counts under each tokenizer and their ratio to the English count for this sentence. The Hausa letter ɗ takes two tokens because N-ATLaS splits it into its two bytes. The sentence was chosen because it is short. It illustrates the comparison and is not a measurement.

4.3 Spelling changes remove very different shares of the excess

Where does the excess come from? We altered the spelling in controlled ways and measured what moved. Figure 5 shows the token reductions, and Table 5 shows what each reduction does to the premium.

Table 5. Token-total premium before and after each alteration, and the share of measured excess removed, with 95% intervals.

Alteration	N-ATLaS premium, before to after	N-ATLaS excess removed (%)	Gemma 4 premium, before to after	Gemma 4 excess removed (%)
Yoruba: all marks removed	2.689 to 1.697	58.72 [58.08, 59.35]	2.370 to 1.582	57.54 [56.86, 58.21]
Yoruba: tone marks removed	2.689 to 1.992	41.28 [40.73, 41.82]	2.370 to 1.771	43.76 [43.16, 44.33]
Yoruba: underdots removed	2.689 to 2.306	22.67 [22.24, 23.10]	2.370 to 2.099	19.80 [19.41, 20.19]
Igbo: all marks removed	2.374 to 2.029	25.12 [24.61, 25.61]	2.140 to 1.852	25.21 [24.71, 25.72]
Hausa: hooked letters replaced	2.021 to 1.914	10.49 [9.99, 10.98]	1.751 to 1.699	6.93 [6.57, 7.31]

Removing all nonspacing marks lowers N-ATLaS's Yoruba token count by 36.88% and removes 58.72%

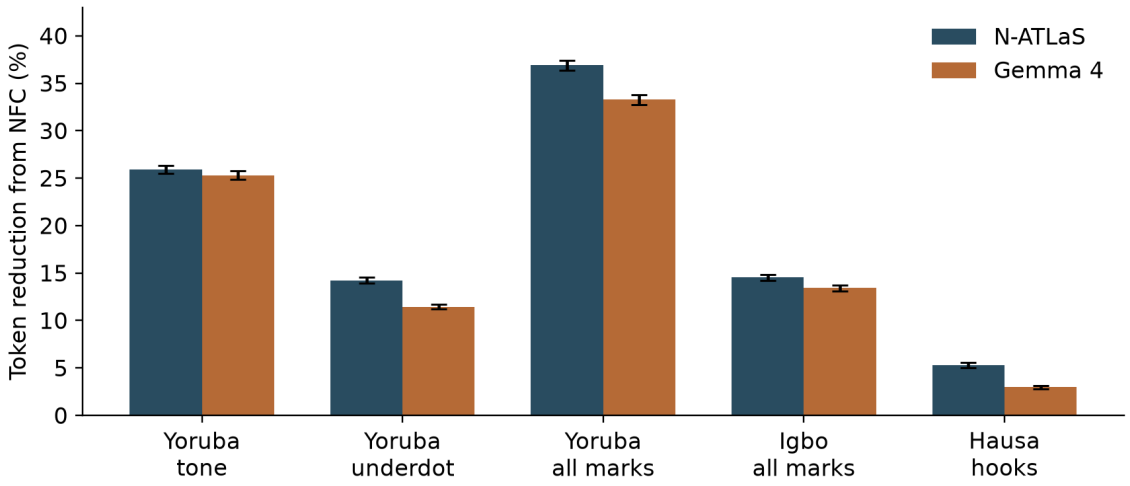


Figure 5. Token-count reductions relative to each sentence's NFC reference. Error bars are bootstrap 95% intervals. "All marks" deletes Unicode category Mn; hooked-letter substitution is a separate Hausa alteration. A reduction does not make the altered spelling an acceptable substitute.

of its measured excess over English. In Igbo the same alteration lowers tokens by 14.54% and removes 25.12% of the excess. Replacing hooked letters in Hausa lowers tokens by 5.30% and removes 10.49%.

The mark inventory explains why the two marked languages behave differently. FLORES+ Yoruba contains 49,323 tone marks and 15,736 underdots. FLORES+ Igbo contains 332 tone marks and 23,745 underdots. So in Igbo, removing tone marks saves only 0.245% of N-ATLaS tokens [0.199, 0.295], while removing underdots saves 14.27% [13.95, 14.59]. Nearly all of Igbo's effect comes from its underdotted letters.

The Yoruba tone and underdot effects cannot be added together. Under N-ATLaS, tone removal saves 36,921 tokens and underdot removal saves 20,279, a sum of 57,200. Removing both at once saves 52,474 tokens, or 36.85%, which is 4,726 fewer than the sum. Removing every Mn mark saves 52,526 tokens, or 36.88%; the extra 52 tokens come from the few marks outside the tone and underdot sets. We therefore report the separate and joint alterations and do not assign additive shares.

Gemma 4's all-mark alterations remove 57.54% of its Yoruba excess and 25.21% of its Igbo excess, close to the N-ATLaS figures. The similarity is descriptive. It does not show that the two tokenizers incur the same orthographic costs for the same reasons.

Hausa is the exception. About 90% of its measured excess remains after the hooked letters are replaced, and this experiment does not identify where the rest comes from.

Table 6. Vocabulary entries that decode to text containing a complete lowercase letter.

Letter	N-ATLaS (128,256 entries)	Gemma 4 (262,144 entries)
ẹ	6	11
ọ	24	36
ṣ	0	3
ị	28	34
ụ	25	29
ƙ	0	2
ɗ	0	1
ɓ	0	1
ƴ	0	1

No individually decoded N-ATLaS entry contains a complete lowercase ṣ, ƙ, ɗ, ɓ or ƴ. These letters are

still encodable, across several entries each. As standalone NFC strings, *ƙarƙashin* takes seven N-ATLaS tokens and five Gemma 4 tokens, and *lọ́wọ́lọ́wọ́* takes twelve under both. Frequent short words also contribute to the total excess, so extreme examples like these should not be read as typical.

Table 7 sets these two words beside a few others. The English words span similar lengths and are not translations. The shortest function words, *da*, *na* and *ní*, cost one token each, the same as English *in*. These three are common words, with 3,795, 2,552 and 1,286 occurrences in the FLORES+ text. The cost appears as soon as a word carries a marked letter. Because no N-ATLaS entry holds a complete ƙ, *ƙasa* takes one token for each of the two bytes that encode the letter and a third for *asa*. In *wọ́n* the acute accent becomes a token of its own, and four code points take four tokens. Marked letters are not the whole explanation, though. The words *muhimmanci* and *gburugburu* contain none, and each takes five N-ATLaS tokens where English *information* takes one. The table illustrates the measurements in Tables 4 and 5. It does not add to them.

Table 7. Example words as standalone NFC strings, with their length in Unicode code points and their token count under each tokenizer. The English words span similar lengths and are not translations of the other words.

Language	Word	Code points	N-ATLaS tokens	Gemma 4 tokens
English	in	2	1	1
English	home	4	1	1
English	school	6	1	1
English	information	11	1	1
Hausa	da	2	1	1
Hausa	ƙasa	4	3	2
Hausa	ɗaya	4	3	2
Hausa	muhimmanci	10	5	4
Igbo	na	2	1	1
Igbo	ndị	3	2	2
Igbo	ahụ	3	2	2
Igbo	gburugburu	10	5	4
Yoruba	ní	2	1	1
Yoruba	àwọn	4	3	3
Yoruba	wọ́n	4	4	3
Yoruba	lọ́wọ́lọ́wọ́	12	12	12

4.4 The Unicode form changes the count without changing the text

Encoding the same Yoruba sentences in NFD instead of NFC raises the token count by 18.74% [18.39, 19.10] under N-ATLaS and by 26.49% [26.02, 26.95] under Gemma 4. For Igbo sentences the increases are 7.99% [7.79, 8.18] and 11.57% [11.30, 11.84]. Hausa barely changes, at 0.02% and 0.03%. Nothing about the text differs between the two forms. The tokenizer simply receives a different sequence of code points.

Because NFD costs Gemma 4 more than it costs N-ATLaS, the comparison between the two tokenizers depends on the form of the input (Figure 6). Gemma 4's Yoruba advantage falls from 11.80% [11.60, 12.00] on NFC to 6.04% [5.91, 6.18] on NFD, and its Igbo advantage from 9.80% to 6.81%. The ranking does not reverse.

These NFD figures come from deliberately converting the whole corpus. The effect on the text as actually published is smaller: normalizing published Yoruba FLORES+ to NFC saves 1,472 N-ATLaS tokens, or 1.02%, and 1,684 Gemma 4 tokens, or 1.32%.

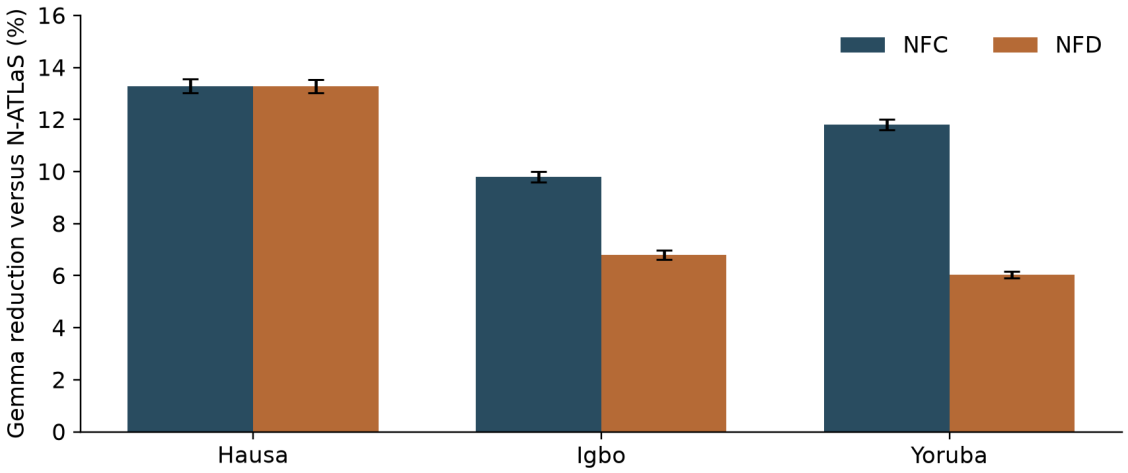


Figure 6. Gemma 4's relative token-count reduction against N-ATLaS on NFC and NFD versions of the same aligned sentences. Error bars are paired-bootstrap 95% intervals.

4.5 The tokenizer difference holds in news and tweets

Gemma 4 uses between 9.20% and 16.99% fewer tokens than N-ATLaS in all nine combinations of language and corpus (Table 8).

Table 8. Gemma 4 token reductions relative to N-ATLaS (%), with 95% intervals. All three corpora use published or reconstructed text, so the FLORES+ column differs slightly from the NFC values in Section 4.2.

Language	FLORES+	News	Tweets
Hausa	13.29 [13.03, 13.55]	10.80 [10.67, 10.94]	16.99 [16.83, 17.15]
Igbo	9.77 [9.56, 9.97]	10.91 [10.84, 10.98]	9.20 [9.01, 9.40]
Yoruba	11.53 [11.32, 11.73]	11.02 [10.90, 11.15]	11.33 [11.21, 11.44]

Could the tweet result be an artefact of URLs and user mentions? Removing both raises Gemma 4's reductions to 18.67% for Hausa, 12.36% for Igbo and 12.24% for Yoruba, so those platform features do not explain the advantage. Removing them changes the strings being measured, and we treat it as a sensitivity check.

For Yoruba, Gemma 4's reduction is 12.10% on the 11,739 tweets that contain combining marks and 6.48% on the 3,151 tweets that contain none. These are different messages, which may differ in length, vocabulary and code-mixing, so the contrast is descriptive only.

The news and tweet results show that the difference between the tokenizers extends beyond aligned translations. They say nothing about the English-relative premium in those registers, because neither corpus has matched English text.

5 Discussion

5.1 Adapting the weights left the representation where it was

The audited N-ATLaS release keeps the complete Llama 3.1 encoding vocabulary. With a fixed tokenizer, updating a model's weights cannot change how a given input is split. So whatever fine-tuning did for N-ATLaS's ability in Hausa, Igbo and Yoruba, it did nothing for what those languages cost to represent.

Why does that matter if the model got better? Because the two properties are paid for in different places. If the community evaluation is right that N-ATLaS outperforms its base on these languages (Ajayi and

Oladipo, 2025), then a Yoruba speaker gets better answers and still fits about 37% as much content into the same token budget as an English speaker. Capability and representation are separate design properties, and we think they should be reported separately. A model card that says a model was adapted for a language should also say whether its tokenizer was.

This is not a criticism particular to N-ATLaS. Fine-tuning or continued pretraining of an existing model is the usual way to build models for African languages (Buzaaba et al., 2025; Yu et al., 2026), and the base tokenizer comes along unless someone decides otherwise. N-ATLaS is a clear case because it was built for exactly the languages that carry the premium.

The premiums themselves describe one aligned corpus. They do not generalize to every Hausa, Igbo or Yoruba request, or to generated answers whose length we did not measure. A tokenizer audit complements downstream evaluation and does not replace it.

5.2 Standard orthography deserves its own evaluation

Underdotted letters and tone marks carry real distinctions in these languages. Our alterations show that token counts drop sharply when some of those distinctions are removed, most of all in Yoruba, where the premium falls from 2.69 to 1.70. The tokenizers are at their most expensive on text written the way the orthography requires.

We want to be careful about what this does and does not show. It does not show that marked spelling causes a fixed fraction of the premium, because deleting marks can produce unfamiliar or ambiguous strings and change segmentation in nonlinear ways. It is not a reason to ask anyone to drop marks or hooked letters. And the unmarked tweets in our sample do not tell us why their writers left the marks out. Keyboard support, convention and register are all possible reasons, and none of them is something we measured.

What the result does support is a change in how these languages are evaluated. Token counts and model performance should be reported on clearly stated orthographic conditions, so that a model that only looks efficient on unmarked text can be told apart from one that handles the standard spelling.

Hausa needs a different explanation. Its hooked letters account for about a tenth of its measured excess, so most of its premium does not come from special characters. We did not test what it does come from.

5.3 NFC is the one change these results let us recommend

NFC and NFD are canonically equivalent, and on our corpora NFC used fewer tokens in aggregate under both tokenizers. Where an application needs the canonical text and not the exact code points, normalizing input to NFC before tokenization is a practical choice that loses no information. Applications that need exact byte or code-point fidelity have to decide separately. We did not measure how normalization affects model quality, and a model trained on mixed forms will see different token sequences after its input is normalized.

The result also matters for measurement. If two studies use different normalization forms, their tokenizer comparisons are not comparable, and a tokenizer's advantage can depend partly on input encoding. We saw this directly: Gemma 4's Yoruba advantage is about twice as large on NFC as on NFD. Benchmarks for marked languages should state the form they use.

5.4 A different tokenizer narrows the gap without closing it

Gemma 4 encodes the measured Nigerian-language text more compactly than N-ATLaS in every corpus we tried. That tells us part of the premium depends on the tokenizer and is not a fixed property of the languages. But Gemma 4 still carries premiums of 1.75 to 2.37 over English, and it removes almost the same share of excess under mark removal as N-ATLaS does.

We cannot say why Gemma 4 does better. Its vocabulary is twice the size, and its training data and segmentation design differ as well. Our experiments cannot separate these factors, and we measured neither the cost of a larger embedding table nor model utility.

5.5 What this means for teams adapting models

For a team building on an existing model, the practical point is that the tokenizer premium can be measured before any training, from tokenizer files alone. It can inform the choice of base model, and it can inform a later decision to adapt the vocabulary. Methods for that exist (Yamaguchi et al., 2024; Ahia et al., 2024; Limisiewicz et al., 2024; Purason et al., 2026), though we tested none of them.

Our vocabulary audit suggests which characters and word forms to look at first. It is not a validated adaptation strategy. The experiment that would turn it into one holds training data and vocabulary capacity constant, introduces targeted vocabulary changes, and measures both the token savings and the downstream effects.

6 Limitations and research ethics

We measure encoding, not outcomes. No model weights, quality benchmarks, inference timings, energy measurements or prices are evaluated. Associations reported in earlier work between fertility and accuracy cannot establish such effects for N-ATLaS.

The premium rests on one translated corpus. FLORES+ is a translated benchmark with its own register and translation process. Aligned translations control the intended content better than unrelated corpora do, but the translators' choices are part of the measured ratio. News and tweets extend the comparison between tokenizers without giving an English baseline. Rebuilding MasakhaNER sentences changes punctuation spacing, and the tweets are released, pre-processed strings and not untouched platform data.

Our units are operational. Whitespace units and offset overlaps are measurement conventions, and the boundary check tests robustness without being a linguistic annotation. Character counts use Unicode code points, so their values depend on normalization.

The perturbations are lossy and they interact. They may produce strings that are not valid words, and their effects are not additive. The share of excess removed describes a text alteration and is not a causal share of the premium.

The Gemma 4 comparison is confounded. Vocabulary size, training data and segmentation design vary together, and one alternative tokenizer is limited evidence. Gemma 4 is also not the most efficient tokenizer reported for these three languages. Somide (2026) finds lower premiums under o200k_base, which we did not measure.

Intervals for natural text assume independence. We cannot cluster by article, author or thread, and removing exact duplicates does not remove all related content. We do not quantify code-mixing.

The identity findings are tied to pinned releases. They apply to the revisions and components we inspected. Matching vocabularies and tokenizer definitions do not prove a shared training history, identical model weights, or equality of every wrapper option. Behavioural equality was observed without chat formatting or added special tokens. The Llama 3.1 behavioural checks cover FLORES+ only; the larger three-corpus check uses Llama 3.

Scope. We study three languages in Latin script and one adapted model. Hausa in Ajami script, other Nigerian languages and other adapted models are outside this study.

Data and ethics. We keep the licence statements of each corpus, report the MasakhaNER licence discrepancy, and publish aggregate results, with no tweet quotations. The FLORES+ dataset card asks that the data not be redistributed publicly without protection from automatic scraping, and that it not

be used as training data. The only corpus text we show is a small number of example words and the one sentence in Figure 4, which is reproduced as an image and credited. The spelling perturbations are analytical diagnostics and not prescriptions for how these languages should be written.

7 Conclusion

N-ATLaS was adapted for Hausa, Igbo and Yoruba, but its vocabulary and the way it splits ordinary text were not. At the audited revisions it shares the complete token-to-ID mapping and loaded tokenizer of Llama 3.1, and the chat template of Llama 3.1 Instruct, with an added padding assignment among the special-token roles we inspected. On aligned NFC text, its Hausa, Igbo and Yoruba token totals are 2.02, 2.37 and 2.69 times the English totals.

That excess responds very differently to spelling in each language. Removing combining marks takes away more than half of Yoruba's measured excess and a quarter of Igbo's, while Hausa's hooked letters account for about a tenth of its excess. The Unicode form matters too: the same Yoruba text costs almost a fifth more tokens in NFD than in NFC, and that roughly halves Gemma 4's advantage on Yoruba. The difference between the two tokenizers holds in news and tweets.

So when a model is adapted for a language, the tokenizer and the input normalization deserve their own audit, next to the usual evaluation of what the model can do. Whether changing the tokenizer would improve an adapted model's quality and its real deployment efficiency is the experiment that should come next.

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Use of AI tools

We used Codex (OpenAI), an AI coding agent, to write and run the experiment code under the author's direction, and Claude (Anthropic) to draft the first version of this paper. The author reviewed the experiments, the results and the text before publication and takes full responsibility for the content.

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Appendix A: Reproducibility and identity details

Table A1. Full resource revisions.

Resource	Revision
N-ATLaS	`e294476928aca9030e924ca27bb8e085e8581273`
Llama 3 8B	`8cde5ca8380496c9a6cc7ef3a8b46a0372a1d920`
Llama 3.1 8B	`d04e592bb4f6aa9cfee91e2e20afa771667e1d4b`
Llama 3.1 8B Instruct	`0e9e39f249a16976918f6564b8830bc894c89659`
Gemma 4 12B	`023679ed352de9bb66cc873c9009ce3482585c08`
FLORES+	`5fec6c13f9e5a4db2f745d4ec0d7c9721ddc4f06`
MasakhaNER 2.0, Hugging Face loader	`60512e89e68841b6b5ed1be59caf97b169f0d27a`
MasakhaNER 2.0, source files	`ba5843cd08aa491d5f96a5e809e71eb9ec461391`
NaijaSenti, Hugging Face loader	`a3d0415a828178edf3466246f49cfcd83b946ab3`
NaijaSenti, source files	`3f267dd565573500ee34e99df19171be3e056be0`

Table A2. SHA-256 of the prepared corpus inputs.

Prepared input	SHA-256
FLORES+	`3be0e3183b9708b3fa2d39f838de5b498d75994f148344e0824c088a32cc2103`
MasakhaNER 2.0 (reconstructed)	`b673146f9201d3975d7691589c8db6d4689f716f39391b374a4c6db1a6b635ad`
NaijaSenti	`856b6eeb68301d6199af05ef62461495ec9c21b0e7661029035eaa05da12517c`

Table A3. SHA-256 of tokenizer source files. Every file differs between N-ATLaS and each Llama 3.1 release.

File	Model	SHA-256
tokenizer.json	N-ATLaS	`6b9e4e7fb171f92fd137b777cc2714bf87d11576700a1dcd7a399e7bbe39537b`
tokenizer.json	Llama 3.1 8B	`76e48799b099d43365bd24ccd8ecc5aedac831718da780552f03b0a6eb4412aa`
tokenizer.json	Llama 3.1 8B Instruct	`79e3e522635f3171300913bb421464a87de6222182a0570b9b2ccba2a964b2b4`
tokenizer_config.json	N-ATLaS	`cf9caa6e3956c24f5787bff455fccb25532afa5091d6b8cbee01a5b7235a1362`
tokenizer_config.json	Llama 3.1 8B	`8004530facf809ac432114de2a4dcc65fcb632da5ec16d666091aeb6a2ee444a`
tokenizer_config.json	Llama 3.1 8B Instruct	`177c7b61e616fecb84c17ce0591acb92c6c4d60e9ac5ababfb940ff23bbcd424`
special_tokens_map.json	N-ATLaS	`1b1835caa5b4d70acaa210fa222b0036f1882f9525c4660fd4810fb3e1e40ff8`
special_tokens_map.json	Llama 3.1 8B	`462d91939dbc37178aa5a3eae7068d1990ccc92e09f288cc71f42cdf139d69cc`
special_tokens_map.json	Llama 3.1 8B Instruct	`6f38c73729248f6c127296386e3cdde96e254636cc58b4169d3fd32328d9a8ec`

Analysis environment. Python 3.11.15, transformers 5.17.0, tokenizers 0.23.2, datasets 5.0.1, huggingface-hub 1.33.0, pandas 3.0.6, numpy 2.4.6, scipy 1.17.1 and matplotlib 3.11.2.

Code and data. The project repository, github.com/prewsh/natlas-token-tax, holds the analysis code and aggregate results. The Llama 3.1 identity reports are `results/tokenizer_identity_llama31.json` and `results/tokenizer_identity_llama31_instruct.json`, and the corpus check is reproduced by `src/complete_llama31_audit.py`. The example words in Table 7 come from `src/example_word_table.py`, with its output in `results/example_words/`, and the sentence in Figure 4 from `src/sentence_examples.py`. These files are at commit `0a7fda9`. Corpus text, raw sentence records and corpus-derived word inventories are not published.

Appendix B: Descriptive metrics and mark inventory

Table B1. Characters per token on NFC text, in Unicode code points.

Language	Tokenizer	With whitespace	Without whitespace
English	N-ATLaS	4.8555	4.0842
Hausa	N-ATLaS	2.5720	2.1286
Igbo	N-ATLaS	2.0767	1.7078
Yoruba	N-ATLaS	1.7281	1.3942
English	Gemma 4	4.8525	4.0818
Hausa	Gemma 4	2.9663	2.4548
Igbo	Gemma 4	2.3024	1.8934
Yoruba	Gemma 4	1.9592	1.5807

Table B2. Secondary fertility premiums and boundary sensitivity on NFC text, with 95% intervals.

Language	Tokenizer	Unit definition	Premium
Hausa	N-ATLaS	Whitespace	1.751 [1.739, 1.763]
Hausa	N-ATLaS	Unicode boundary	1.753 [1.741, 1.765]
Hausa	Gemma 4	Whitespace	1.517 [1.508, 1.527]
Hausa	Gemma 4	Unicode boundary	1.519 [1.510, 1.528]
Igbo	N-ATLaS	Whitespace	2.103 [2.087, 2.119]
Igbo	N-ATLaS	Unicode boundary	2.045 [2.030, 2.061]
Igbo	Gemma 4	Whitespace	1.895 [1.880, 1.911]
Igbo	Gemma 4	Unicode boundary	1.844 [1.830, 1.858]
Yoruba	N-ATLaS	Whitespace	2.326 [2.299, 2.351]
Yoruba	N-ATLaS	Unicode boundary	2.334 [2.308, 2.359]
Yoruba	Gemma 4	Whitespace	2.050 [2.027, 2.072]
Yoruba	Gemma 4	Unicode boundary	2.057 [2.035, 2.079]

Table B3. Fragmentation of whitespace units on NFC FLORES+ text (% of units).

Language	Tokenizer	1 token	2 tokens	3 tokens	4 or more
English	N-ATLaS	82.89	13.36	2.61	1.14
Hausa	N-ATLaS	34.02	35.11	20.14	10.73
Igbo	N-ATLaS	23.91	35.99	20.55	19.56
Yoruba	N-ATLaS	26.89	27.66	17.67	27.78
English	Gemma 4	84.18	11.79	2.54	1.50
Hausa	Gemma 4	44.11	34.23	16.03	5.64
Igbo	Gemma 4	34.97	28.49	18.23	18.31
Yoruba	Gemma 4	34.26	24.69	19.72	21.33

Table B4. Token change relative to NFC for each alteration (%), with 95% intervals. Reductions for the spelling alterations, increases for NFD.

Language	Alteration	N-ATLaS	Gemma 4
Yoruba	Tone marks removed	25.93 [25.49, 26.35]	25.30 [24.84, 25.74]
Yoruba	Underdots removed	14.24 [13.93, 14.54]	11.45 [11.18, 11.71]
Yoruba	All marks removed	36.88 [36.33, 37.42]	33.27 [32.73, 33.81]
Igbo	Tone marks removed	0.245 [0.199, 0.295]	0.212 [0.170, 0.255]
Igbo	Underdots removed	14.27 [13.95, 14.59]	13.20 [12.90, 13.49]
Igbo	All marks removed	14.54 [14.21, 14.86]	13.43 [13.12, 13.73]
Hausa	Hooked letters replaced	5.30 [5.04, 5.57]	2.97 [2.81, 3.14]
Hausa	NFD (increase)	0.0215 [0.0094, 0.0362]	0.0312 [0.0151, 0.0494]
Igbo	NFD (increase)	7.99 [7.79, 8.18]	11.57 [11.30, 11.84]
Yoruba	NFD (increase)	18.74 [18.39, 19.10]	26.49 [26.02, 26.95]

Table B5. Combining marks in FLORES+ after decomposition. "Covered" means the mark belongs to the tone or underdot sets.

Language	Code point	Combining mark	Count	Covered
Hausa	U+0301	Acute accent	17	Yes
Hausa	U+0303	Tilde	3	No
Hausa	U+0304	Macron	1	Yes
Hausa	U+0306	Breve	1	No
Hausa	U+0308	Diaeresis	6	No
Hausa	U+0327	Cedilla	2	No
Igbo	U+0300	Grave accent	251	Yes
Igbo	U+0301	Acute accent	59	Yes
Igbo	U+0303	Tilde	13	No
Igbo	U+0304	Macron	22	Yes
Igbo	U+0306	Breve	1	No
Igbo	U+0307	Dot above	5	No
Igbo	U+0308	Diaeresis	7	No
Igbo	U+0323	Dot below	23,745	Yes
Igbo	U+0326	Comma below	1	No
Igbo	U+0327	Cedilla	2	No
Yoruba	U+0300	Grave accent	23,139	Yes
Yoruba	U+0301	Acute accent	26,184	Yes
Yoruba	U+0302	Circumflex accent	1	No
Yoruba	U+0303	Tilde	2	No
Yoruba	U+0308	Diaeresis	12	No
Yoruba	U+0323	Dot below	15,736	Yes
Yoruba	U+0327	Cedilla	1	No
Yoruba	U+0328	Ogonek	30	No

U+0329 (vertical line below) does not occur in any of the three languages in this corpus.

Appendix C: Supporting significance tests

All tests are approximate two-sided Wilcoxon signed-rank tests on paired token counts with `zero_method='zsplit'`. SciPy records the smallest values as zero through numerical underflow. We display those as below 1e-300, and they should not be read as exact zero probabilities.

Table C1. Orthographic alterations and NFD against the NFC reference. Holm correction covers all 34 comparisons in the focused orthography run, including auxiliary conditions not shown.

Language	Tokenizer	Alteration	Sentences with fewer / equal / more tokens	Holm-adjusted p
Hausa	N-ATLaS	Hooked letters replaced	1,143 / 866 / 0	9.29e-223
Hausa	Gemma 4	Hooked letters replaced	1,072 / 937 / 0	2.74e-207
Hausa	N-ATLaS	NFD	0 / 1,995 / 14	1
Hausa	Gemma 4	NFD	0 / 1,994 / 15	1
Igbo	N-ATLaS	Tone marks removed	132 / 1,877 / 0	3.24e-07
Igbo	Gemma 4	Tone marks removed	135 / 1,871 / 3	3.24e-07
Igbo	N-ATLaS	Underdots removed	1,796 / 213 / 0	below 1e-300
Igbo	Gemma 4	Underdots removed	1,793 / 216 / 0	below 1e-300
Igbo	N-ATLaS	All marks removed	1,797 / 212 / 0	below 1e-300
Igbo	Gemma 4	All marks removed	1,794 / 215 / 0	below 1e-300
Igbo	N-ATLaS	NFD	0 / 235 / 1,774	below 1e-300
Igbo	Gemma 4	NFD	0 / 222 / 1,787	below 1e-300
Yoruba	N-ATLaS	Tone marks removed	1,656 / 353 / 0	below 1e-300
Yoruba	Gemma 4	Tone marks removed	1,655 / 354 / 0	below 1e-300
Yoruba	N-ATLaS	Underdots removed	1,690 / 319 / 0	below 1e-300
Yoruba	Gemma 4	Underdots removed	1,684 / 324 / 1	below 1e-300
Yoruba	N-ATLaS	All marks removed	1,705 / 304 / 0	below 1e-300
Yoruba	Gemma 4	All marks removed	1,702 / 306 / 1	below 1e-300
Yoruba	N-ATLaS	NFD	2 / 313 / 1,694	below 1e-300
Yoruba	Gemma 4	NFD	0 / 307 / 1,702	below 1e-300

Table C2. Gemma 4 against N-ATLaS on published or reconstructed text. Holm correction covers four languages for FLORES+ and three languages for each natural corpus.

Corpus	Language	Texts where Gemma 4 uses fewer / equal / more tokens	Holm-adjusted p
FLORES+	English	623 / 855 / 531	0.0673
FLORES+	Hausa	1,933 / 43 / 33	below 1e-300
FLORES+	Igbo	1,917 / 47 / 45	below 1e-300
FLORES+	Yoruba	1,935 / 46 / 28	below 1e-300
News	Hausa	7,261 / 347 / 305	below 1e-300
News	Igbo	9,922 / 401 / 58	below 1e-300
News	Yoruba	9,211 / 391 / 31	below 1e-300
Tweets	Hausa	20,807 / 1,012 / 118	below 1e-300
Tweets	Igbo	11,666 / 3,028 / 983	below 1e-300
Tweets	Yoruba	13,693 / 847 / 350	below 1e-300

Appendix D: Terms used in this paper

Term	Meaning in this paper
Token	A unit of text that a language model reads or writes. A word can be one token or several.
Tokenizer	The component that splits text into tokens, using a fixed vocabulary.
Vocabulary	The fixed list of tokens a tokenizer knows. Each token has an ID number.
Token-to-ID mapping	The full table that pairs every token in the vocabulary with its ID.
Special or control token	A reserved token that marks structure, such as the start of text, the end of a turn or padding. It does not represent ordinary text.
Chat template	The fixed text pattern a model expects around a conversation, built from special tokens.
Code point	The number Unicode assigns to a character or mark, written as U+ followed by a hexadecimal value.
Combining mark	A code point that attaches to the letter before it, such as a tone mark or a dot below.
NFC	Unicode Normalization Form C. A letter and its marks are stored as one combined code point wherever one exists.
NFD	Unicode Normalization Form D. A letter and its marks are stored separately, the plain letter first and each mark after it.
Tone mark	An accent that shows tone. In this study: the grave, acute and macron accents.
Underdot	A dot below a letter, as in ẹ, ọ and ṣ in Yoruba and ị, ọ and ụ in Igbo.
Hooked letter	The Hausa letters ɓ, ɗ, ƙ and ƴ.
Fertility	Tokens per whitespace-delimited unit of text.
Token-total premium	The total tokens a language needs for the aligned sentences, divided by the total English needs for the same sentences.
Fertility premium	A language's fertility divided by English fertility under the same tokenizer.
Tokenizer tax	A name used in recent work for the extra tokens a language needs over English. It refers to the same extra cost that this paper measures as a premium.
Excess	The part of a premium above 1, which is the part above the English level.
Share of excess removed	How much of that excess disappears when the text is altered in a specified way.
Fragmentation	The share of whitespace units that are split into one, two, three, or four or more tokens.
Bootstrap confidence interval	A range for an estimate, obtained by recomputing it on many random resamples of the sentences. We report 95% intervals.